

## Annex 31B

(normative)

### MAC Control PAUSE operation

#### 31B.1 PAUSE description

The PAUSE operation is used to inhibit transmission of data frames for a specified period of time. A MAC Control client wishing to inhibit transmission of data frames from another station on the network generates a `MA_CONTROL.request` primitive specifying the following:

- a) The globally assigned 48-bit multicast address 01-80-C2-00-00-01
- b) The PAUSE opcode
- c) A `request_operand` indicating the length of time for which it wishes to inhibit data frame transmission (see 31B.2)

The PAUSE operation cannot be used to inhibit transmission of MAC Control frames.

PAUSE frames shall only be sent by DTEs configured to the full duplex mode of operation.

The globally assigned 48-bit multicast address 01-80-C2-00-00-01 has been reserved for use in MAC Control PAUSE frames for inhibiting transmission of data frames from a DTE in a full duplex mode IEEE 802.3 LAN. IEEE 802.1Q-conformant bridges will not forward frames sent to this multicast destination address, regardless of the state of the bridge's ports, or whether or not the bridge implements the MAC Control sublayer. To allow generic full duplex flow control, stations implementing the PAUSE operation shall instruct the MAC (e.g., through layer management) to enable reception of frames with destination address equal to this multicast address.

NOTE—By definition, an IEEE 802.3 LAN operating in full duplex mode comprises exactly two stations, thus there is no ambiguity regarding the destination DTE's identity. The use of a well-known multicast address relieves the MAC Control sublayer and its client from having to know, and maintain knowledge of, the individual 48-bit address of the other DTE in a full duplex environment.

When MAC Control PFC operation (see Annex 31D and IEEE Std 802.1Q) has been enabled, MAC Control PAUSE operation shall be disabled.

#### 31B.2 Parameter semantics

The PAUSE opcode takes the following `request_operand`:

`pause_time`

A 2-octet, unsigned integer containing the length of time for which the receiving station is requested to inhibit data frame transmission. The field is transmitted most-significant octet first, and least-significant octet second. The `pause_time` is measured in units of `pause_quanta`, equal to 512 bit times of the particular implementation (see 4.4). The range of possible `pause_time` is 0 to 65535 `pause_quanta`.

### 31B.3 Detailed specification of PAUSE operation

The state machines in 31B.3 follow the conventions in 21.5. (See 25.1.1 for an example statement. That also covers the timer conventions from 14 which apply here.)

#### 31B.3.1 Transmit operation

Upon receipt of a MA\_CONTROL.request primitive containing the PAUSE opcode from a MAC client, the MAC Control sublayer calls the MAC sublayer MAC:MA\_DATA.request service primitive with the following parameters:

- a) The destination\_address is set equal to the destination\_address parameter of the MA\_CONTROL.request primitive. This parameter is currently restricted to the value specified in 31B.1.
- b) The source\_address is set equal to the 48-bit individual address of the station.
- c) The length/type field (i.e., the first two octets) within the mac\_service\_data\_unit parameter is set to the IEEE 802.3 MAC Control EtherType value assigned in 31.4.1.3.
- d) The remainder of the mac\_service\_data\_unit is set equal to the concatenation of the PAUSE opcode encoding (see Annex 31A), the pause\_time request\_operand specified in the MA\_CONTROL.request primitive, and a field containing zeros of the length specified in 31.4.1.6.
- e) The frame\_check\_sequence is omitted.

Upon receipt of a data transmission request from the MAC client through the MCF:MA\_DATA.request primitive, if the transmission of data frames has not been inhibited due to reception of a valid MAC Control frame specifying the PAUSE operation and a non-zero pause\_time, the MAC Control sublayer calls the MAC sublayer MAC:MA\_DATA.request service primitive with its parameters identical to the MCF:MA\_DATA.indication primitive.

#### 31B.3.2 Transmit state diagram for PAUSE operation

It is not required that an implementation be able to transmit PAUSE frames. MAC Control sublayer entities that transmit PAUSE frames shall implement the Transmit state diagram specified in this subclause.

##### 31B.3.2.1 Constants

pause\_command

The 2-octet encoding of the PAUSE command, as specified in Annex 31A.

reserved\_multicast\_address

The 48-bit address specified in 31B.1 item a).

802.3\_MAC\_Control

The 16 bit EtherType field assignment for IEEE 802.3 MAC Control specified in 31.4.1.3.

phys\_Address

A 48-bit value, equal to the unicast address of the station implementing the MAC Control sublayer.

##### 31B.3.2.2 Variables

transmitEnabled

A Boolean set by network management to indicate that the station is permitted to transmit on the network.

Values: true; Transmitter is enabled by management

false; Transmitter is disabled by management

transmission\_in\_progress

A Boolean used to indicate to the Receive state diagram that a MAC:MA\_DATA.request service primitive call is pending.

Values: true; transmission is in progress  
false; transmission is not in progress

n\_quanta\_tx

An integer indicating the number of pause\_quanta that the transmitter of a PAUSE frame is requesting that the receiver pause.

pause\_timer\_Done

A Boolean set by the MAC Control PAUSE Operation Receive state diagram to indicate the expiration of a pause\_timer initiated by reception of a MAC Control frame with a PAUSE opcode.

Values: true; The pause\_timer has expired.  
false; The pause\_timer has not expired.

pause\_mac\_service\_data\_unit

The concatenation of IEEE 802.3\_MAC\_Control, pause\_command, n\_quanta\_tx, zeros.

### 31B.3.2.3 Functions

None

### 31B.3.2.4 Timers

pause\_timer

The timer governing the inhibition of transmission of data frames (see 31.5.1) from a MAC Client by the PAUSE function.

### 31B.3.2.5 Messages

MA\_CONTROL.request

The service primitive used by a client to request a MAC Control sublayer function with the specified request\_operands.

MCF:MA\_DATA.request

The service primitive used by a client to request MAC data transmission with the specified parameters.

MAC:MA\_DATA.request

The service primitive issued by the MAC Control sublayer to transmit a MAC frame with the specified parameters. The action invoked is not considered to end until the transmission of the frame by the MAC has concluded and the ability of the MAC control layer to determine this is implementation dependent.

### 31B.3.2.6 Transmit state diagram for PAUSE operation

Figure 31B–1 depicts the Transmit state diagram for a MAC Control sublayer entity implementing the PAUSE operation.

## 31B.3.3 Receive operation

The opcode-independent MAC Control sublayer Receive state diagram accepts and parses valid frames received from the MAC sublayer. MAC Control sublayer entities that implement the PAUSE operation shall implement the Receive state diagram specified in this subclause. The functions specified in this subclause are performed upon receipt of a valid Control frame containing the PAUSE opcode, and define the function called by the INITIATE MAC CONTROL FUNCTION state of Figure 31–4. (See 31.5.3.)

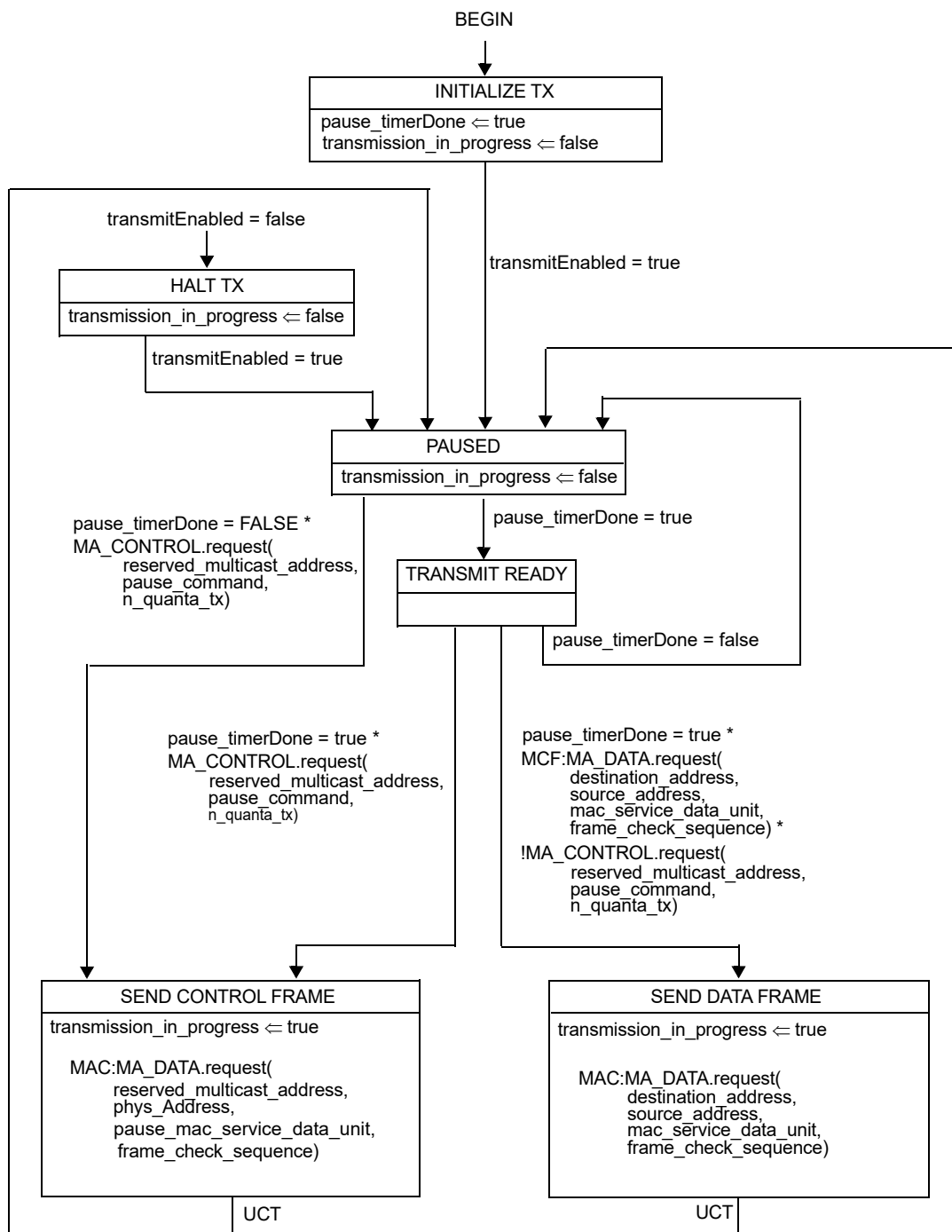


Figure 31B-1—PAUSE Operation Transmit state diagram

Upon receipt of a valid MAC Control frame with the opcode indicating PAUSE and the destination address indicating either: (1) the globally assigned multicast address specified in 31B.1, or (2) the unique physical address associated with this station, the MAC Control sublayer starts a timer for the length of time specified by the `pause_time request_operand` in the Control frame (see 31B.2). The value of the variable `pause_timer_Done` is set to false upon the setting of the `pause_timer` to a non-zero value by the MAC Control PAUSE Operation Receive state diagram. The value of the variable `pause_timer_Done` is set to true when the timer value reaches zero (i.e., the timer expires). If the received PAUSE operation indicates a `pause_timer` value of zero, the value of `pause_timer_Done` is set to true immediately.

The receipt of a valid MAC Control frame with the opcode indicating PAUSE and the destination address indicating the globally assigned multicast address specified in 31B.1 or the unique physical address associated with this station will cause the `pause_timer` to be set according to the `pause_time request_operand` in the newly received frame, regardless of the current setting of the `pause_timer`, i.e., new PAUSE operations override earlier PAUSE operations.

### 31B.3.4 Receive state diagram for PAUSE operation

#### 31B.3.4.1 Constants

- `pause_quantum`  
The unit of measurement for pause time specified in 31B.2
- `pause_command`  
The 2-octet encoding of the PAUSE command, as specified in Annex 31A.
- `reserved_multicast_address`  
The 48-bit address specified in 31B.1 item a).
- `phys_Address`  
A 48-bit value, equal to the unicast address of the station implementing the MAC Control sublayer.

#### 31B.3.4.2 Variables

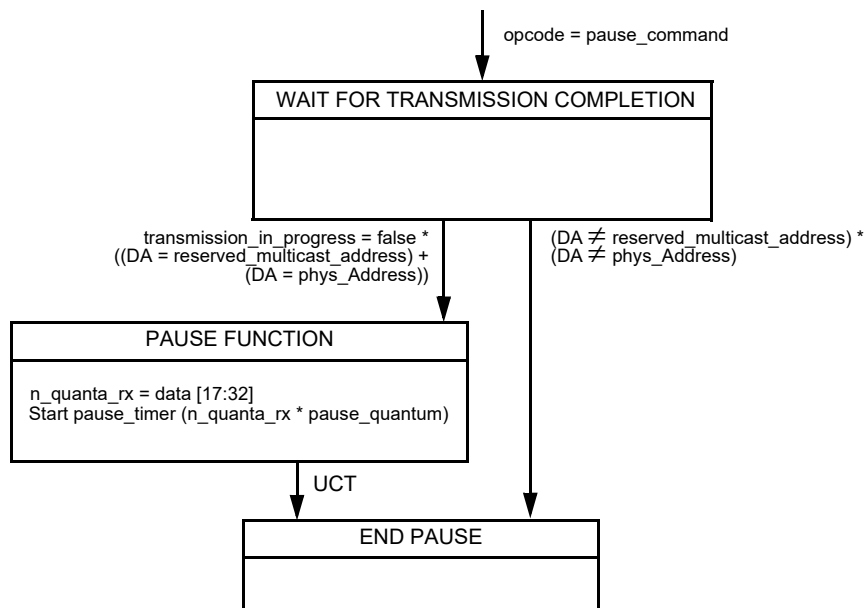
- `n_quanta_rx`  
An integer used to extract the number of `pause_quanta` to set the `pause_timer` from the `dataParam` of the received MAC frame.
- `opcode`  
The opcode parsed from the received MAC frame.
- `DA`  
The destination address from the `MAC:MA_DATA.indication` service primitive.
- `data`  
The data payload parsed from the received MAC frame.
- `transmission_in_progress`  
A Boolean used by the Transmit state diagram to indicate that a `MAC:MA_DATA.request` service primitive call is pending.  
Values: true; transmission is in progress  
false; transmission is not in progress

#### 31B.3.4.3 Timers

- `pause_timer`  
The timer governing the inhibition of transmission of data frames.

### 31B.3.4.4 Receive state diagram (INITIATE MAC CONTROL FUNCTION) for PAUSE operation

Figure 31B–2 depicts the INITIATE MAC CONTROL FUNCTION for the PAUSE operation. (See 31.5.3.)



**Figure 31B–2—PAUSE Operation Receive state diagram**

### 31B.3.5 Status indication operation

MAC Control sublayer entities that implement the PAUSE operation shall implement the Indication state diagram specified in this subclause.

The PAUSE function sets the `pause_status` variable to one of two values: `paused` or `not_paused`, and indicates this to its client through the `MA_CONTROL.indication` primitive.

### 31B.3.6 Indication state diagram for pause operation

#### 31B.3.6.1 Constants

`pause_command`

The 2-octet encoding of the PAUSE command, as specified in Annex 31A.

#### 31B.3.6.2 Variables

`pause_status`

Used to indicate the state of the MAC Control sublayer for the PAUSE operation. It takes one of two values: `paused` or `not_paused`.

`pause_timer_Done`

A Boolean set by the MAC Control PAUSE Operation Receive state diagram to indicate the expiration of a `pause_timer` initiated by reception of a MAC Control frame with a PAUSE opcode.

Values: `true`; The `pause_timer` has expired.

`false`; The `pause_timer` has not expired.

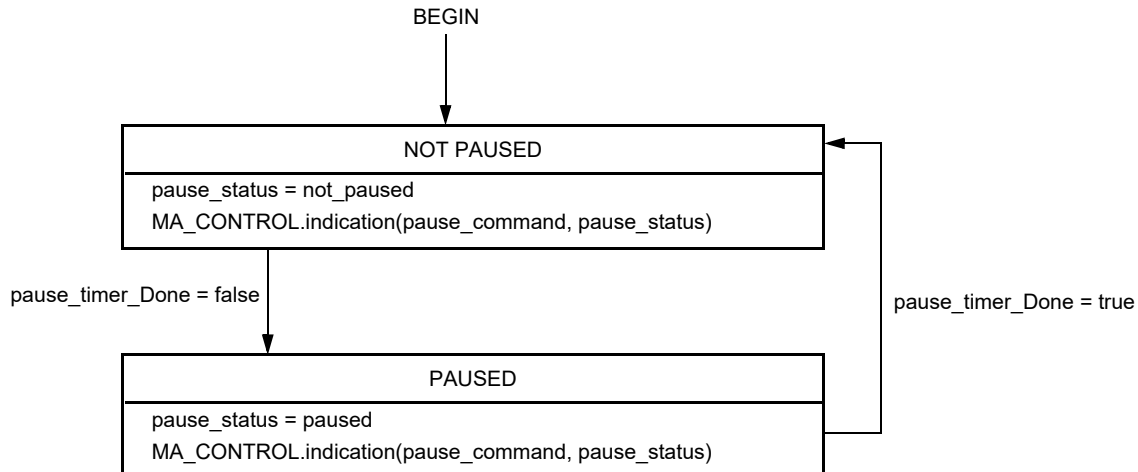
### 31B.3.6.3 Messages

MA\_CONTROL.indication

The service primitive used to indicate the state of the MAC Control sublayer to its client.

### 31B.3.6.4 Indication state diagram for PAUSE operation

Figure 31B–3 depicts the Indication State Diagram for a MAC Control sublayer implementing the PAUSE operation.



**Figure 31B–3—PAUSE Operation Indication state diagram**

### 31B.3.7 Timing considerations for PAUSE operation

In a full duplex mode DTE, it is possible to receive PAUSE frames asynchronously with respect to the transmission of MAC frames. For effective flow control, it is necessary to place an upper bound on the length of time that a DTE can transmit data frames after receiving a valid PAUSE frame with a non-zero pause\_time request\_operand.

Reception of a PAUSE frame shall not affect the transmission of a MAC frame that has been submitted by the MAC Control sublayer to the underlying MAC (i.e., the MAC:MA\_DATA.request service primitive is synchronous, and is never interrupted).

At operating speeds of 100 Mb/s or less, a station that implements an exposed MII, shall not begin to transmit a (new) frame (assertion of TX\_EN at the MII, see 22.2.2.3) more than one pause\_quantum after the reception of a valid PAUSE frame (deassertion of RX\_DV at the MII, see 22.2.2.7) that contains a nonzero value of pause\_time. Stations that do not implement an exposed MII, shall measure this time at the MDI, with the timing specification increased to one pause\_quantum + 64 BT.

At an operating speed of 1000 Mb/s, a station shall not begin to transmit a (new) frame more than two pause\_quanta after the reception of a valid PAUSE frame that contains a non-zero value of pause\_time, as measured at the MDI.

At operating speeds of 2.5 Gb/s, a station with a 2.5GBASE-T PHY shall not begin to transmit a (new) frame more than 34 pause\_quanta after the reception of a valid PAUSE frame that contains a non-zero value of pause\_time, as measured at the MDI. A station using any other 2.5 Gb/s PHY shall not begin to transmit a (new) frame more than 6 pause\_quanta after the reception of a valid PAUSE frame that contains a non-zero value of pause\_time, as measured at the MDI.

At operating speeds of 5 Gb/s, a station with a 5GBASE-T PHY shall not begin to transmit a (new) frame more than 42 pause\_quanta after the reception of a valid PAUSE frame that contains a non-zero value of pause\_time, as measured at the MDI. A station using any other 5 Gb/s PHY shall not begin to transmit a (new) frame more than 12 pause\_quanta after the reception of a valid PAUSE frame that contains a non-zero value of pause\_time, as measured at the MDI.

At operating speeds of 10 Gb/s, a station with a 10GBASE-T PHY shall not begin to transmit a (new) frame more than 74 pause\_quanta after the reception of a valid PAUSE frame that contains a non-zero value of pause\_time, as measured at the MDI. A station using any other 10 Gb/s PHY shall not begin to transmit a (new) frame more than 60 pause\_quanta after the reception of a valid PAUSE frame that contains a non-zero value of pause\_time, as measured at the MDI.

At operating speeds of 25 Gb/s, a station shall not begin to transmit a (new) frame more than 80 pause\_quanta after the reception of a valid PAUSE frame that contains a non-zero value of pause\_time, as measured at the MDI.

At operating speeds of 40 Gb/s, a station shall not begin to transmit a (new) frame more than 118 pause\_quanta after the reception of a valid PAUSE frame that contains a non-zero value of pause\_time, as measured at the MDI.

At operating speeds of 50 Gb/s, a station shall not begin to transmit a (new) frame more than 117 pause\_quanta after the reception of a valid PAUSE frame that contains a non-zero value of pause\_time, as measured at the MDI.

At operating speeds of 100 Gb/s, a station shall not begin to transmit a (new) frame more than 394 pause\_quanta after the reception of a valid PAUSE frame that contains a non-zero value of pause\_time, as measured at the MDI.

At operating speeds of 200 Gb/s, a station shall not begin to transmit a (new) frame more than 453 pause\_quanta after the reception of a valid PAUSE frame that contains a non-zero value of pause\_time, as measured at the MDI.

At operating speeds of 400 Gb/s, a station shall not begin to transmit a (new) frame more than 905 pause\_quanta after the reception of a valid PAUSE frame that contains a non-zero value of pause\_time, as measured at the MDI.

In addition to DTE and MAC Control delays, system designers should take into account the delay of the link segment when designing devices that implement the PAUSE operation.

The PAUSE response time may be verified by demonstrating that no more than max\_overrun bytes of frame data are sent by the station after reception of a valid PAUSE frame that contains a non-zero value of pause\_time. The value of max\_overrun is defined for the following operating speeds, where frame\_length is the maximum frame length transmitted by the station during the test:

2.5 Gb/s (using 2.5GBASE-T) –  $\text{max\_overrun} = 2176 + \text{frame\_length}$ .

2.5 Gb/s (not using 2.5GBASE-T) –  $\text{max\_overrun} = 384 + \text{frame\_length}$

5 Gb/s (using 5GBASE-T) –  $\text{max\_overrun} = 2624 + \text{frame\_length}$ .

5 Gb/s (not using 5GBASE-T) –  $\text{max\_overrun} = 768 + \text{frame\_length}$

10 Gb/s (using 10GBASE-T) –  $\text{max\_overrun} = 4736 + \text{frame\_length}$ .

10 Gb/s (not using 10GBASE-T) –  $\text{max\_overrun} = 3840 + \text{frame\_length}$ .

25 Gb/s – max\_overshoot = 5120 + frame\_length.

40 Gb/s – max\_overshoot = 7552 + frame\_length.

50 Gb/s – max\_overshoot = 7488 + frame\_length.

100 Gb/s – max\_overshoot = 25216 + frame\_length.

200 Gb/s – max\_overshoot = 28992 + frame\_length.

400 Gb/s – max\_overshoot = 57920 + frame\_length.

## 31B.4 Protocol implementation conformance statement (PICS) proforma for MAC Control PAUSE operation<sup>272</sup>

### 31B.4.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Annex 31B, MAC Control PAUSE operation, shall complete the following PICS proforma in addition to the PICS of Clause 31.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

### 31B.4.2 Identification

#### 31B.4.2.1 Implementation identification

|                                                                                                                                                                                                                                                                                                                                            |  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Supplier                                                                                                                                                                                                                                                                                                                                   |  |
| Contact point for inquiries about the PICS                                                                                                                                                                                                                                                                                                 |  |
| Implementation Name(s) and Version(s)                                                                                                                                                                                                                                                                                                      |  |
| Other information necessary for full identification—<br>e.g., name(s) and version(s) for machines and/or<br>operating systems; System Name(s)                                                                                                                                                                                              |  |
| <p>NOTE 1—Only the first three items are required for all implementations, other information may be completed as appropriate in meeting the requirements for the identification.</p> <p>NOTE 2—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).</p> |  |

#### 31B.4.2.2 Protocol summary

|                                                                                                                                                                            |                                                             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| Identification of protocol standard                                                                                                                                        | IEEE Std 802.3-2022, Annex 31B, MAC Control PAUSE operation |
| Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS                                                            |                                                             |
| <p>Have any Exception items been required? No [ ] Yes [ ]</p> <p>(See Clause 21; The answer Yes means that the implementation does not conform to IEEE Std 802.3-2022)</p> |                                                             |

|                   |  |
|-------------------|--|
| Date of Statement |  |
|-------------------|--|

<sup>272</sup>Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this annex so that it can be used for its intended purpose and may further publish the completed PICS.

### 31B.4.3 Major capabilities/options

| Item   | Feature                                                                                               | Subclause | Value/Comment | Status | Support           |
|--------|-------------------------------------------------------------------------------------------------------|-----------|---------------|--------|-------------------|
| *PST   | Support for transmit of PAUSE frames                                                                  | 31B.3.2   | N/A           | O      | Yes [ ]<br>No [ ] |
| *MIIa  | At operating speeds of 100 Mb/s or less, MII connection exists and is accessible for test.            | 31B.3.7   | N/A           | O      | Yes [ ]<br>No [ ] |
| *MIIb  | At operating speeds of 100 Mb/s or less, MII connection does not exist or is not accessible for test. | 31B.3.7   | N/A           | O      | Yes [ ]<br>No [ ] |
| *MIIc  | At operating speeds of 1000 Mb/s                                                                      | 31B.3.7   | N/A           | O      | Yes [ ]<br>No [ ] |
| *MIId  | At operating speeds of 2.5 Gb/s with PHY types other than 2.5GBASE-T.                                 | 31B.3.7   | N/A           | O      | Yes [ ]<br>No [ ] |
| *MIIe  | At operating speeds of 2.5 Gb/s with PHY types of 2.5GBASE-T.                                         | 31B.3.7   | N/A           | O      | Yes [ ]<br>No [ ] |
| *MIIf  | At operating speeds of 5 Gb/s with PHY types other than 5GBASE-T.                                     | 31B.3.7   | N/A           | O      | Yes [ ]<br>No [ ] |
| *MIIfg | At operating speeds of 5 Gb/s with PHY types of 5GBASE-T.                                             | 31B.3.7   | N/A           | O      | Yes [ ]<br>No [ ] |
| *MIIh  | At operating speeds of 10 Gb/s with PHY types other than 10GBASE-T.                                   | 31B.3.7   | N/A           | O      | Yes [ ]<br>No [ ] |
| *MIIi  | At operating speeds of 10 Gb/s with PHY types of 10GBASE-T.                                           | 31B.3.7   | N/A           | O      | Yes [ ]<br>No [ ] |
| *MIIj  | At operating speeds of 25 Gb/s.                                                                       | 31B.3.7   | N/A           | O      | Yes [ ]<br>No [ ] |
| *MIIk  | At operating speeds of 40 Gb/s.                                                                       | 31B.3.7   | N/A           | O      | Yes [ ]<br>No [ ] |
| *MIIl  | At operating speeds of 50 Gb/s.                                                                       | 31B.3.7   | N/A           | O      | Yes [ ]<br>No [ ] |
| *MIIm  | At operating speeds of 100 Gb/s.                                                                      | 31B.3.7   | N/A           | O      | Yes [ ]<br>No [ ] |
| *MIIn  | At operating speeds of 200 Gb/s.                                                                      | 31B.3.7   | N/A           | O      | Yes [ ]<br>No [ ] |
| *MIIo  | At operating speeds of 400 Gb/s.                                                                      | 31B.3.7   | N/A           | O      | Yes [ ]<br>No [ ] |

### 31B.4.4 PAUSE command requirements

| Item | Feature                                                                                       | Subclause | Value/Comment | Status    | Support |
|------|-----------------------------------------------------------------------------------------------|-----------|---------------|-----------|---------|
| PCR1 | Duplex mode of DTE                                                                            | 31B.1     | Full duplex   | M         | Yes [ ] |
| PCR2 | Reception of frames with destination address equal to the multicast address 01-80-C2-00-00-01 | 31B.1     | Enabled       | M         | Yes [ ] |
| PCR3 | PAUSE operation on PFC enable                                                                 | 31B.1     | Disabled      | PFC:<br>M | Yes [ ] |

### 31B.4.5 PAUSE command state diagram requirements

| Item | Feature                                      | Subclause | Value/Comment                      | Status | Support               |
|------|----------------------------------------------|-----------|------------------------------------|--------|-----------------------|
| PSD1 | Transmit state diagram for PAUSE operation   | 31B.3.2   | Meets requirements of Figure 31B–1 | PST: M | N/A [ ]<br>M: Yes [ ] |
| PSD2 | Receive state diagram for PAUSE operation    | 31B.3.4   | Meets requirements of Figure 31B–2 | M      | Yes [ ]               |
| PSD3 | Indication state diagram for PAUSE operation | 31B.3.6   | Meets requirements of Figure 31B–3 | M      | Yes [ ]               |

### 31B.4.6 PAUSE command MAC timing considerations

| Item | Feature                                                                       | Subclause | Value/Comment                                              | Status | Support            |
|------|-------------------------------------------------------------------------------|-----------|------------------------------------------------------------|--------|--------------------|
| TIM1 | Effect of PAUSE frame on a frame already submitted to underlying MAC          | 31B.3.7   | Has no effect                                              | M      | Yes [ ]            |
| TIM2 | Measurement point for station with MII                                        | 31B.3.7   | Delay at MII $\leq$ one pause_quantum <sup>a</sup>         | MIIf:M | Yes [ ]<br>N/A [ ] |
| TIM3 | Measurement point for station without MII at 100 Mb/s or less                 | 31B.3.7   | Delay at MDI $\leq$ one pause_quantum + 64 BT <sup>a</sup> | MIIf:M | Yes [ ]<br>N/A [ ] |
| TIM4 | Measurement point for station at 1000 Mb/s                                    | 31B.3.7   | Delay at MDI $\leq$ two pause_quanta <sup>a</sup>          | MIIf:M | Yes [ ]<br>N/A [ ] |
| TIM5 | Measurement point for station at 2.5 Gb/s with PHY type other than 2.5GBASE-T | 31B.3.7   | Delay at MDI $\leq$ 6 pause_quanta <sup>a</sup>            | MIIf:M | Yes [ ]<br>N/A [ ] |
| TIM6 | Measurement point for station at 2.5 Gb/s with PHY type of 2.5GBASE-T         | 31B.3.7   | Delay at MDI $\leq$ 34 pause_quanta <sup>a</sup>           | MIIf:M | Yes [ ]<br>N/A [ ] |
| TIM7 | Measurement point for station at 5 Gb/s with PHY type other than 5GBASE-T     | 31B.3.7   | Delay at MDI $\leq$ 12 pause_quanta <sup>a</sup>           | MIIf:M | Yes [ ]<br>N/A [ ] |

| Item  | Feature                                                                      | Subclause | Value/Comment                                     | Status  | Support            |
|-------|------------------------------------------------------------------------------|-----------|---------------------------------------------------|---------|--------------------|
| TIM8  | Measurement point for station at 5 Gb/s with PHY type of 5GBASE-T            | 31B.3.7   | Delay at MDI $\leq$ 42 pause_quanta <sup>a</sup>  | MIlg:M  | Yes [ ]<br>N/A [ ] |
| TIM9  | Measurement point for station at 10 Gb/s with PHY types other than 10GBASE-T | 31B.3.7   | Delay at MDI $\leq$ 60 pause_quanta <sup>a</sup>  | MIlh:M  | Yes [ ]<br>N/A [ ] |
| TIM10 | Measurement point for station at 10Gb/s with PHY type of 10GBASE-T           | 31B.3.7   | Delay at MDI $\leq$ 74 pause_quanta <sup>a</sup>  | MIli:M  | Yes [ ]<br>N/A [ ] |
| TIM11 | Measurement point for station at 25 Gb/s                                     | 31B.3.7   | Delay at MDI $\leq$ 60 pause_quanta <sup>a</sup>  | MIlj:M  | Yes [ ]<br>N/A [ ] |
| TIM12 | Measurement point for station at 40 Gb/s                                     | 31B.3.7   | Delay at MDI $\leq$ 118 pause_quanta <sup>a</sup> | MIlk:M  | Yes [ ]<br>N/A [ ] |
| TIM13 | Measurement point for station at 50 Gb/s                                     | 31B.3.7   | Delay at MDI $\leq$ 118 pause_quanta <sup>a</sup> | MIll:M  | Yes [ ]<br>N/A [ ] |
| TIM14 | Measurement point for station at 100 Gb/s                                    | 31B.3.7   | Delay at MDI $\leq$ 117 pause_quanta <sup>a</sup> | MIlm:M  | Yes [ ]<br>N/A [ ] |
| TIM15 | Measurement point for station at 200 Gb/s                                    | 31B.3.7   | Delay at MDI $\leq$ 453 pause_quanta <sup>a</sup> | MIln: M | Yes [ ]<br>N/A [ ] |
| TIM16 | Measurement point for station at 400 Gb/s                                    | 31B.3.7   | Delay at MDI $\leq$ 905 pause_quanta <sup>a</sup> | MIlo: M | Yes [ ]<br>N/A [ ] |

<sup>a</sup>Delay from receiving valid PAUSE command, with nonzero value for pause\_time, to cessation of transmission.